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Maya: An Instruction Finetuned Multilingual Multimodal Model

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Abstract

The rapid development of large Vision-Language Models (VLMs) has led to impressive results on academic benchmarks, primarily in widely spoken languages. However, significant gaps remain in the ability of current VLMs to handle low-resource languages and varied cultural contexts, largely due to a lack of high-quality, diverse, and safety-vetted data. Consequently, these models often struggle to understand low-resource languages and cultural nuances in a manner free from toxicity. To address these limitations, we introduce Maya, an open-source Multimodal Multilingual model. Our contributions are threefold: 1) a multilingual image-text pretraining dataset in eight languages, based on the LLaVA pretraining dataset; 2) a thorough analysis of toxicity within the LLaVA dataset, followed by the creation of a novel toxicity-free version across eight languages; and 3) a multilingual image-text model supporting these languages, enhancing cultural and linguistic comprehension in vision-language tasks. Code available at https://github.com/nahidalam/maya.

1. Introduction

Vision Language Models (VLMs) have emerged as a key technique in artificial intelligence, enabling machines to understand and reason about the visual world through natural language. Recent advancements in Large Language Models (LLMs) and general-purpose image encoders such as CLIP [44] and SigLIP [63] has significantly boosted VLM capabilities. Architectures such as Flamingo [5], LLaVA [34, 35], KOSMOS [40, 42], Florence-2 [59] and Molmo [16] demonstrate strong performance across tasks including image captioning, Visual Question Answering (VQA), and complex reasoning. Qwen2-VL [58] introduced Multimodal Rotary Position Embedding (M-RoPE) [50] and dynamic resolution techniques, while PaLI's joint modality scaling and cross-lingual learning [13, 14] have furthered vision-language understanding. Despite these advances, VLMs remain limited to high-resource languages. This creates a disparity in accessibility, as current models struggle with cultural contexts and visual concepts in low-resource languages [26].

This gap is largely due to the lack of high-quality multilingual multimodal datasets. Pre-training datasets like COCO [31], Flickr30K [61], LAION [46], Visual Genome [27], and LLaVA [35] are predominantly in English, limiting cross-linguistic generalization. Existing multilingual datasets, such as Multi30k [18] and Crossmodal-3600 [56], are limited in scale and cultural diversity. Current datasets often contain toxic and culturally insensitive content [62], perpetuating biases and stereotypes. To our knowledge, no peer-reviewed research has systematically addressed the mitigation of toxicity and harmful content in image-text datasets. This highlights the need for comprehensive, toxicity-mitigated datasets that support robust, multilingual training while capturing linguistic and cultural diversity.

To address these challenges, we introduce Maya, an open-source Multilingual Multimodal Vision Language Model (mVLM) that expands multimodal capabilities to

*Work does not relate to the authors' positions.

eight languages with an emphasis on data quality and cultural sensitivity. Built on the LLaVA framework, Maya includes a newly created pre-training dataset designed to support multilingual and culturally aware VLM development. Our key contributions include:

1. a novel multilingual image-text pretraining dataset consisting of *558,000* images for future development of mVLMs,
2. a toxicity-free version of the dataset in 8 languages, and
3. a new mVLM that demonstrates improved performance in understanding cultural and linguistic nuances compared to PALO-7B [37] on LLaVA-Bench-In-The-Wild [35], offering a multilingual alternative to LLaVA [35].

2. Related Work

2.1. Multilingual Large Language Models

While open-source large language models have made notable strides, progress in multilingual LLMs (mLLMs) still lags. Many LLMs [1, 4, 17, 54] offer limited multilingual support, typically focusing on high-resource languages with reduced performance in others. Dedicated mLLMs [7, 9, 28, 30, 32, 57, 60], such as BLOOM [28] and Aya-101 [57], extend language coverage to 43 and 101 languages respectively, improving performance across tasks like translation, summarization, and reasoning.

2.2. Multimodal Large Language Models

Research in Multimodal Large Language Models (MLLMs) has advanced significantly in enabling models to process both images and text across diverse tasks, with progress along two main paths: proprietary closed-source systems [6, 39, 53] and open-source initiatives. While most of the initial efforts focused predominantly on English-language capabilities, such as LLaVA [34, 35], several recent works have attempted to close this gap by targeting non-English languages. PALO [37] extended LLaVA [35] to cover 10 major languages via semi-automated translations of the LLaVA-Instruct-150k dataset [35]. X-LLaVA [48] demonstrated strong Korean-English bilingual performance. Broader language coverage has been achieved by Llama 3.2 [17] supporting 8 languages and Qwen2-VL [58] covering several European languages alongside Japanese, Korean, Arabic and Vietnamese. More recent developments have pushed multilingual boundaries through diverse approaches: PaliGemma [10] pretrained on a variety of vision-language tasks upto 35 languages providing an open base MLLM but without instruction tuning, [19] mBLIP realigned English-tuned image encoder using machine-translated data for 95 languages, Pangea [62] is a multilingual multimodal model trained on an instruction dataset spanning 39 languages, Parrot [51] addressed supervised fine-tuning imbalances across 6 languages through textual guidance for visual token alignment, while models like Phi-3.5-Vision [1], Pixtral [3] and Molmo [16] are mostly English-centric [62].

2.3. Multimodal Multilingual Datasets

Large-scale datasets for retrieval and captioning such as xFlickrCO [12], BLIP3-KALE [8] and WIT [49] have been developed to support broader multimodal research. For instruction tuning, PALO [37] built a dataset of 150K images in 10 languages from a subset of LLaVA image-text instruction tuning dataset on English text, while Pangea [62] demonstrated broader coverage with image-text pair dataset in 39 languages.

2.4. Evaluating Multimodal Models

Evaluating multilingual multimodal LLMs [10, 58, 62] involves two main categories. The first assesses standard multimodal tasks, such as cross-lingual visual question answering (VQA) with xGQA [43] and image-text retrieval via xFlickrCO [12]. The second evaluates cultural understanding, using datasets like MaRVL [33] for visual reasoning, CVQA [45] for cultural context, and XM3600 [56] for cross-modal comprehension. Significant progress has been made in developing multilingual multimodal datasets for evaluating VLMs including XVNLI [12] for visual natural language inference, MTVQA [52] for temporal reasoning, M3Exam [64] for educational assessment, PangeaBench [62] for broad multilingual capabilities, MMMB [51] for cross-lingual understanding, and LLaVA-Bench-In-The-Wild [35, 37] for real-world scenarios. Notably, none of these datasets focus on evaluating toxicity.

3. Dataset Creation and Filtering

3.1. Dataset Creation Methodology

Recently, PALO [37] and Pangea [62] have created multilingual image-text dataset for building multilingual multimodal models. However, these multilingual datasets often suffer from data quality issues and distribution biases across languages. For instance, in the PALO dataset, the distribution varies significantly between English and other languages [18, 24, 37]. To address these limitations, we present a novel pre-training dataset tailored for LLaVA's architecture that is both multilingual and optimized for diverse language representation. Our dataset introduces rigorous processes for toxicity analysis, distribution balancing, and quality control, ensuring consistent and reliable performance across multiple languages and modalities as shown in Figure 1. We expand the original English LLaVA dataset, which contains 558,000 samples, to include seven additional languages—Chinese, French, Spanish, Russian, Hindi, Japanese, and Arabic—yielding a total of 4.4 million samples, equally distributed across all eight languages.

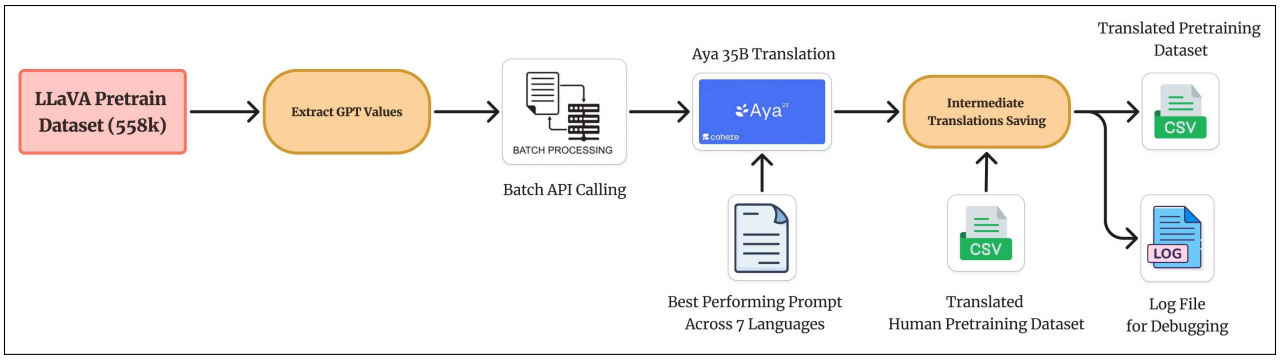

Figure 1. Pretrain Dataset Preparation Process

Every sample in our dataset has undergone toxicity verification and filtering, ensuring a safer training corpus than the original LLaVA pretraining dataset.

Our approach encompasses three core components: 1) parallel dataset creation using a hybrid translation method, 2) prompt engineering optimization, and 3) scalable generation of pre-training datasets. This pipeline integrates multiple language models, such as [6, 39, 53], alongside specialized multilingual models like Aya 35B [7], to ensure high-quality, cross-lingual data suitable for multilingual applications.

3.1.1. Initial Dataset Processing

Our pipeline is built on the LLaVA dataset [35], centered around image-text pairs and their corresponding GPT responses. We use stratified sampling to select 30 representative samples per language, optimizing for linguistic diversity through Length Analysis (LA), Flesch Reading Ease (FRE), and Flesch-Kincaid Grade Level (FKGL) metrics [55]. To ensure quality, we implement a cascaded translation verification system. First, we use Google Translate to do initial translation. This is followed by back-translation verification. Finally human reviewers verify it with the help of [2, 6, 53].

3.1.2. Prompt Engineering and Evaluation

In the prompt engineering phase, we employ a prompt evaluation process for each language. First, we create a prompt evaluation dataset using a similar process as Figure 1. We take 6 sample prompts and translate the English text of the prompt evaluation dataset to seven languages using Aya 35B. We then compare these translations with the reference translation of the prompt evaluation dataset using BLEU score [41] and N-gram score [11, 47]. In Figure 2, we plot the n-gram BLEU scores (averaged across 1-gram, 2-gram, 3-gram, and 4-gram) for each language and preamble type. We find that across all seven languages (Arabic, Chinese, French, Hindi, Japanese, Russian, and Spanish), Preamble 6 consistently achieves the highest N-gram BLEU scores, with most languages showing a notable upward trend from Type 5 to Type 6, reaching values around 0.4-0.5. The radar plot in Figure 3 provides complementary evidence by breaking down the N-gram analysis (1-gram through 4-gram) averaged across all languages for each preamble type. It reveals that Preamble 6 maintains the largest area coverage, indicating superior performance across all N-gram levels compared to other preamble types. This is particularly significant as higher N-gram BLEU scores suggest better preservation of phrase structures and contextual meaning in the translations. The relatively balanced performance of Preamble 6 across different languages and across different N-gram levels indicates its robustness and language-agnostic effectiveness in the translation task. We find that preamble 6 is the best preamble across all the seven languages. The resulting prompt template is in listing 1. We use this prompt template in our translation framework.

Listing 1. Translation Instructions

```
## Instructions
You are an expert in translations.
Your job is to translate the input to Japanese in
the given chat.

Ensure that:
{Specific Things to Consider while Translating}

Note: {Extra Constraints on Output Generation}

## Examples
### Example 1
Input:
{Input Sentence}
Expected Output:
{Translated Sentence}
```

3.1.3. Translation Framework Design

Our translation framework uses the best preamble identified in the prompt engineering evaluation step. The prompt includes 1) standardized input-output formatting to maintain uniformity across languages, 2) example-driven instruction

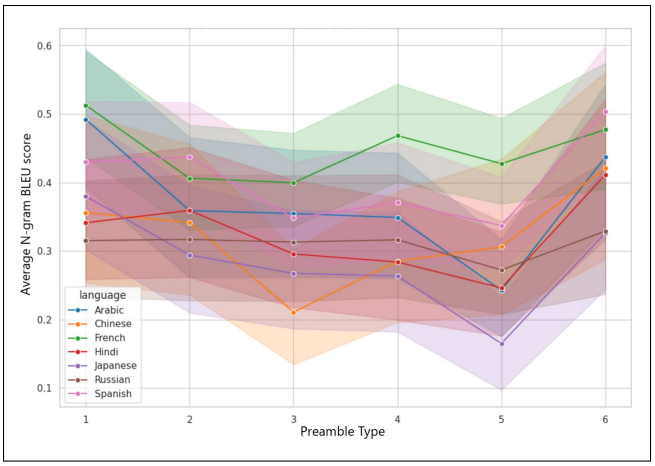

Figure 2. N-gram values by language and preamble type. These values are average of 1-gram, 2-gram, 3-gram and 4-gram

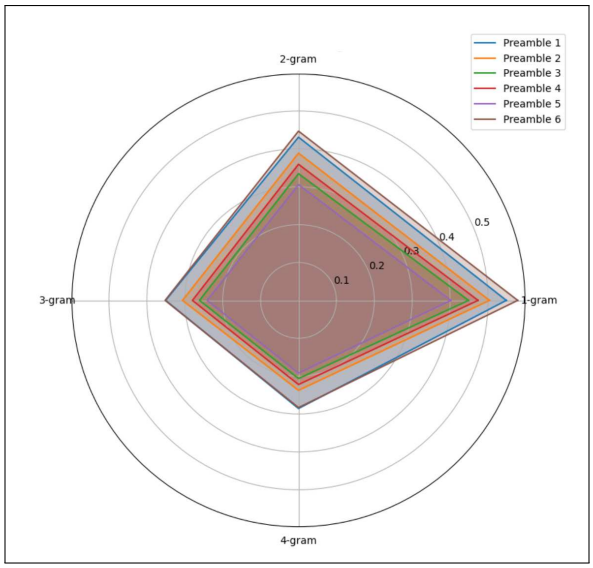

Figure 3. Radar chart of N-gram averages by preamble

sets tailored for complex translations, and 3) integrated validation triggers to ensure quality control. Leveraging Aya 35B advanced translation capabilities, our framework achieves greater than 0.47 average BLEU scores across seven languages.

3.1.4. Scalable Dataset Generation

For large-scale dataset generation, we implement a batch processing pipeline integrated with Aya 35B API as shown in Figure 1. First, we extract GPT values with quality filter from the LLaVA pretrain dataset. These extracted GPT values are then passed through the Aya 35B batch-optimized API calling with intermediate translation checkpointing. The pipeline does necessary error handling and comprehensive logging for quality tracking. This architecture enables efficient processing of the full 558K samples while maintaining translation quality. We implement version control for intermediate translations and maintain detailed debug logs, ensuring reproducibility and enabling systematic error analysis.

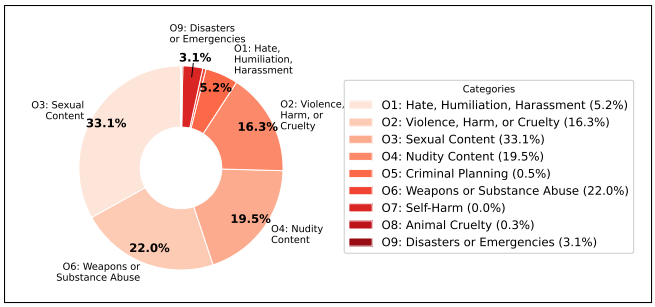

Figure 4. Image Toxicity Analysis with LLaVAGuard

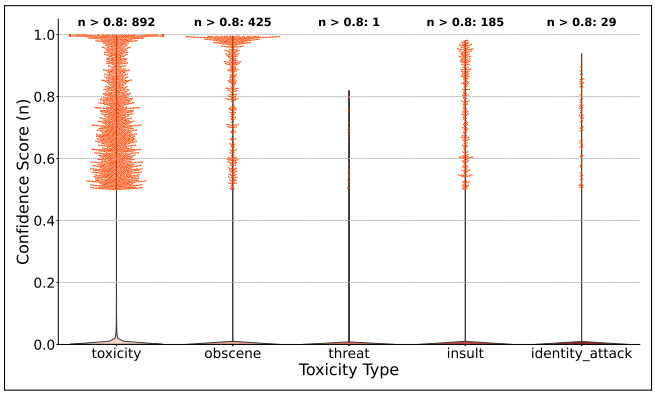

Figure 5. Image Caption Toxicity Analysis with Toxic-BERT

3.2. Dataset Toxicity Filtering

In the original English LLaVA Pretrain dataset which has 558,000 image-caption pairs, we observed some toxic contents. To methodically identify toxic content, we used LLaVAGuard 7B [22] and Toxic-BERT [20]. For images, we relied on the LlavaGuard 7B framework to spot and categorize unsafe or toxic visuals based on set guidelines. For text, we used the Toxic-BERT model to scan captions and flag anything with offensive or harmful language.

LLaVAGuard is specifically designed for in-depth toxicity analysis, assigning safety ratings, categories, and rationales to flagged images, producing structured data about each identified risk. This analysis led to the categorization of content across various types, including sexual content, hate speech, violence, and substance abuse, which are visually represented in the toxicity analysis chart as shown in Figure 4. Additionally, a prompt optimization pipeline was introduced to ensure precise filtering, capturing potentially harmful content while reducing false positives.

The overall process for creating toxicity free dataset is

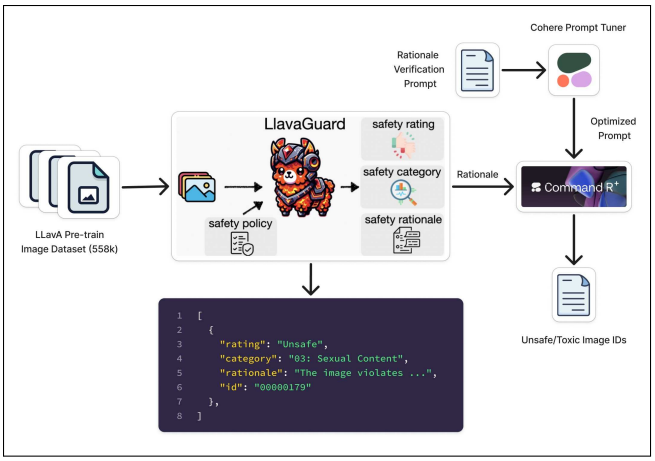

Figure 6. Dataset Toxicity Filtering Method

shown in Figure 6. LLaVaGuard output provides a rating, category, and rationale explaining why an image violates guidelines. We then refine this by identifying genuinely toxic images. This is done by developing an optimized prompt using the Cohere prompt tuner*. We use this prompt as a preamble (System Prompt) and pair it with the previous information—rating, category, and rationale—associated with each image flagged as unsafe by LLaVAGuard. Command R+ [15] then analyzes these results to identify the truly unsafe image IDs. In our analysis, LLaVAGuard identified 7,600 images as toxic and the final Command R+ output identified 7,111 images as unsafe.

For identifying toxic captions, we used Toxic-BERT that categorized 892 image captions as toxic with greater than 80% confidence as shown in Figure 5. LLaVAGuard and Command R+ identified 7,111 images and Toxic-BERT identified 892 images, there are in total 7,531 unique toxic images. We then removed those 7,531 images from the pretraining dataset to create the toxicity free pretraining dataset.

4. Multilingual Multimodal Instruction Tuning

4.1. Model Architecture

The Maya model architecture draws inspiration from LLaVA 1.5 [34]. Our goal is to leverage pretrained multilingual Aya model [7] and vision encoder with multilingual capabilities. Specifically, we employ the Aya-23 8B model as our LLM f_ϕ because of its multilingual capability. Aya-23 has 8 billion parameters, an 8K context window, and is trained across 23 languages. Our dataset, however, includes 8 of these 23 languages, aligning with our objective of optimizing Maya for a diverse yet focused linguistic range.

For the vision encoder, we opted for SigLIP† [63] rather than CLIP [44], which is traditionally used in LLaVA. This choice is motivated by SigLIP's strong performance, multilingual adaptability, and capacity for variable-length patch sizes. Unlike CLIP, SigLIP supports scalable positional embeddings, enabling it to accept inputs of varying dimensions through positional embedding interpolation. This flexibility makes SigLIP particularly suitable for our purpose. For each input image X_v, we get the visual features from SigLIP, $Z_v = g(X_v)$. A trainable projection matrix W then converts the image features Z_v into language features H_v.

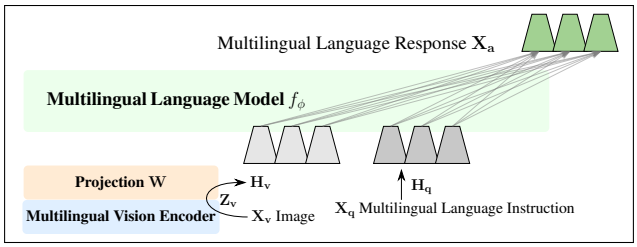

Figure 7. Maya Architecture adapted from LLaVA [36]

4.2. Pretraining

For image-text alignment, we used a projection matrix W that brings image features X_v closer to language features. This projection matrix is a simple 2-layer MLP with GELU activation [23], as in LLaVA 1.5 [34]. Although we experimented with 4- and 8-layer MLPs, the 2-layer configuration consistently achieved the lowest training loss. Advanced alignment techniques, such as gated soft-attention in Flamingo [5], Q-Former in BLIP-2 [29], or pooling from MM1 [38] as alternatives to the projection layer, are set aside for future work. For each image X_v, we used the multi-turn conversation data

$$(X_q^1, X_a^1, \ldots, X_q^T, X_a^T)$$

where T is the total number of turns from LLaVA [35]. We translated this data to 7 languages. We pretrained Maya on this dataset and the toxicity free version of the same. Image inputs were cropped to 256x256 for compatibility with the SigLIP encoder. Training was conducted on 8xH100 GPUs with 80GB DRAM, using a per-device batch size of 32 and a global batch size of 256. A learning rate of 1e-3 and a cosine scheduler were applied during training. The pretraining process only trains the projection matrix and took about 20 hours.

4.3. Finetuning

We instruction-finetuned our pretrained Maya model using the PALO 150K instruction-tuning dataset [37]. During finetuning, we observed that Low Rank Adaptation (LoRA)

*https://docs.cohere.com/v2/docs/prompt-tuner

†siglip-base-patch16-256-multilingual from https://huggingface.co/google/siglip-base-patch16-256-multilingual

[25] produced suboptimal results, particularly when both adapter matrices A and B were updated with the same learning rate [21]. Based on these findings, we opted against using LoRA and instead implemented full finetuning on 8xH100 GPUs. Our finetuning configuration used a per-device batch size of 16 and a global batch size of 128. Finetuning process took about 48 hours.

We kept both the vision encoder and the language encoders frozen during the training process. We did pretraining and finetuning for both versions of our dataset: the pretraining dataset translated into 7 languages, with and without toxicity filtering. This resulted in two variants of our model: Maya and Maya-Toxicity-Free.

5. Results

We evaluate Maya on PALO multilingual evaluation set [37] as shown in Table 1. Although our pretraining dataset only contains eight languages, we finetuned Maya on PALO instruction tuning dataset that contains ten languages. Therefore in Table 1 Maya evaluation results are shown in all of those ten PALO instruction-tuning languages. Maya offers good performance across languages compared to the models within the size class of 7B models, and offers comparable performance to the 13B size models. On average across all languages, Maya outperforms LLaVA-13B and is only slightly worse than PALO-13B. Specifically, among the common eight languages, Maya performs better than PALO 7B in five languages. We think this is due to the fact that Maya pretrain dataset is multilingual in nature whereas PALO pretrain dataset is same as the English-only LLaVA pretrain dataset. Maya also beats PALO and LLaVA in Arabic language for both the size classes of 7B and 13B. The root based language system of Arabic along with the quality of translation due to the choice of preamble led to this result. Another observation we have with PALO multilingual evaluation is that the scores for **Maya** and **Maya-Toxicity Free** are exactly the same. We empirically conclude that this is because multilingual evaluation does not account for content toxicity.

In Figure 8, we compare Maya responses with LLaVA and GPT4. We observe that Maya responses are very similar to LLaVA and lacks the similar level of details vs. GPT4. For example, in answering **Question 1**, both Maya and LLaVA lacks the details with respect to **a plastic box**. When Maya is wrong, for example in **Question 2**, LLaVA is also wrong; although in a different way. The right answer here is **Fage** from GPT4. Both Maya and LLaVA get it wrong saying **Yoplait** and **Chobani** respectively. Additional qualitative evaluation results are added in the supplemental materials.

To compare the effect of filtering the pretraining dataset for toxicity, we evaluate Maya and Maya-Toxicity Free models across various English-only benchmarks as shown in Table 2. Both models have comparable accuracy across most of the benchmarks, with only slight variations. This suggests that removing toxic content from the training data does not significantly degrade overall performance for these tasks. Maya-Toxicity Free shows marginal gains in TextVQA, Text Translation, and Numerical Calculation benchmarks. Performance in Commonsense Reasoning and MM-VeT decreases, indicating that some complex reasoning tasks may benefit from the presence of diverse (and possibly toxic) content in training data.

In Table 3, we show how Maya compares with its toxicity free variant in various VizWiz categories. Maya-

Close-Up of Multiple Everyday Objects

Question 1:
What fruit is in the left part of the fridge?

Maya:
There are strawberries in the left part of the fridge.
LLaVA-7B:
There are strawberries in the left part of the fridge.
GPT4:
In the left part of the fridge, there is a plastic box of strawberries.

Question 2:
What is the brand of the yogurt flavored with blueberry?

Maya:
The brand of the yogurt flavored with blueberry is Yoplait.
LLaVA-7B:
The brand of the yogurt flavored with blueberry is Chobani.
GPT4:
The brand of the blueberry-flavored yogurt is Fage.

Figure 8. Comparison of model responses to visual questions on everyday objects

Model	English	Chinese	French	Spanish	Russian	Japanese	Arabic	Hindi	Bengali	Urdu	Avg.
Maya (8B)	61.5	61.7	61.0	60.4	62.2	63.7	**63.4**	64.0	50.8	55	60.4
LLaVA-7B	67.9	55.7	62.4	64.5	55.3	59.2	38.9	29.4	13.9	21.8	46.9
PALO-7B	64.2	55.7	58.3	61.0	57.4	57.5	57.8	57.6	51.7	55.3	57.7
	-6.4	+6.0	-1.4	-4.1	+4.8	+6.2	+5.6	+6.4	-0.9	-0.3	+2.7
LLaVA-13B	**69.5**	**62.9**	**67.5**	64.6	62.3	**65.3**	37.2	27.8	20.4	22.1	49.9
PALO-13B	65.5	62.1	66.4	**65.9**	**62.4**	60.6	56.9	**66.8**	**53.5**	**59.6**	**61.9**
	-8.0	-1.2	-6.5	-5.5	-0.2	-1.6	+6.5	-2.8	-2.7	-4.6	-1.5

Table 1. A comparison of LLaVA and PALO with Maya on eight languages adapted from LLaVA-Bench (In-the-Wild). Values underlined indicate best performance within size class and values in **bold** indicate best performance across all models tested. We provide performance differences between Maya and the best competing model within the size classes where red indicates where Maya is performing better and blue indicates where Maya is performing worse than the best in the size class. "Avg." represents the average over all the languages.

Benchmarks	**Maya**	**Maya-Toxicity Free**
GQA	57.79%	57.84%
VizWiz	34.92%	34.98%
ScienceQA	70.27%	69.51%
TextVQA	47.01%	48.56%
POPE-adversarial	81.00%	80.97%
POPE-popular	84.10%	82.84%
POPE-random	85.30%	84.27%
MMBench	71.10%	71.12%
MM-VeT	29.8	27.7
MME (Perception + Cognition)	72.45%	71.90%
MME (Cognition):		
Commonsense Reasoning	75.71%	68.57%
Numerical Calculation	47.50%	50.00%
Text Translation	52.50%	55.00%
Code Reasoning	60.00%	57.50%

Table 2. Accuracy of Maya models on English Language across multiple benchmarks.

VizWiz	**Maya**	**Maya-Toxicity Free**
Overall	**34.92%**	**34.98%**
other	34.03%	33.66%
unanswerable	30.88%	32.03%
yes/no	77.02%	77.08%
number	24.63%	24.72%

Table 3. Detailed accuracy results for VizWiz.

Toxicity Free slightly outperforms Maya by 0.06%. This indicates that the removal of toxic content in training data has a minimal positive impact on the overall performance of the VizWiz benchmark. Maya performs slightly better in the "other" category. This suggests that for questions that fall outside of well-defined categories, the unfiltered training data may offer a slight advantage. Maya-Toxicity Free shows improved performance (32.03% vs. 30.88%), suggesting that a cleaner training dataset helps the model better recognize when a question cannot be answered. Both models perform almost identically in the yes/no category indicating that toxicity removal does not significantly affect binary question accuracy.

In Table 4, we show a detailed performance analysis of the Maya and Maya-Toxicity Free models on the MMVeT benchmark, breaking down their overall accuracy, individual capabilities, and more complex capability integrations. In categories like **Spatial Awareness, OCR, Knowledge** and **Spatial Awareness, Language Generation, OCR**, Maya-Toxicity Free shows zero performance, suggesting areas where toxicity removal could have eliminated beneficial patterns. For **Spatial Awareness, OCR, Recognition, Math**, both models perform poorly, indicating the difficulty of this capability combination, with no difference between the models. Overall, MMVeT results highlight that while Maya-Toxicity Free shows some improvements in specific areas, like language generation and certain integrations involving spatial awareness, it generally suffers a performance decline, especially in math, OCR, and integrated recognition tasks. The impact of removing toxic content seems to weaken the model's overall ability to perform complex reasoning and integrate multiple capabilities effectively. We leave out further investigation on this topic for future work.

Figure 10 shows visual question output in different languages for figure 9. We asked Maya **please describe the food in {language} in 1 sentence**. We observe that Bengali reply in this case is more detailed than English, identifying meat on the bowl and wooden table. Spanish, French

MMVeT	Maya	Maya-Toxicity Free
Overall	**29.8**	**27.7**
Capabilities		
Recognition	33.7	32.7
OCR	23.1	18.7
Knowledge	16.0	13.7
Language Generation	18.4	19.3
Spatial Awareness	29.0	26.7
Math	15.4	7.7
Capability Integrations		
Language Generation, Recognition, Knowledge	15.5	16.5
Recognition	69.2	68.9
Spatial Awareness, OCR	25.8	25.4
Spatial Awareness, OCR, Math	28.6	14.3
Spatial Awareness, Recognition	41.7	50.0
OCR	25.8	18.3
OCR, Math	0.0	0.0
Recognition, Knowledge	16.7	6.0
Spatial Awareness, Language Generation, OCR, Recognition	44.2	49.8
Language Generation, OCR, Recognition, Knowledge	15.2	9.0
Spatial Awareness, OCR, Recognition	14.3	14.3
OCR, Recognition	53.0	25.0
Spatial Awareness, OCR, Knowledge	3.3	0.0
Spatial Awareness, Recognition, Knowledge	50.0	0.0
Spatial Awareness, Language Generation, OCR	19.0	24.0
Spatial Awareness, OCR, Recognition, Math	0.0	0.0

Table 4. MMVeT results including individual and integrated capabilities. Values are mean of 5 runs with 0.2 standard deviation.

Figure 9. Example image from LLaVA-Bench (In-the-Wild) [35].

English	A bowl of noodles with vegetables in it.
Spanish	Un plato de fideos asiáticos con verduras y carne.
French	Un bol de nouilles asiatiques avec des légumes et de la viande.
Chinese	一个装满面条和蔬菜的碗，可能是一碗汤面。
Japanese	麺が野菜と一緒に入ったボウルで、おそらくラーメンです。
Russian	Миска с лапшой, возможно, раменом, с различными овощами сверху.
Arabic	وعاء من النودلز مع مكونات مختلفة، ربما حساء.
Hindi	एक कटोरा जिसमें नूडल्स होते हैं, जो संभवतः एक एशियाई सूप होता है, जिसमें सब्जियाँ और मांस होते हैं।
Bengali	একটি বাটি নুডলস সুপ যা সবজি এবং মাংস দিয়ে পূর্ণ করা হয়েছে একটি কাঠের টেবিলে রাখা হয়েছে।
Urdu	یک باؤل نوڈلز جس میں سبزیوں کے ساتھ ہے اور ایک چمچ کے ساتھ ہے۔

Figure 10. Maya output for prompt (with image from Figure 9): Please describe the food in {language} in 1 sentence.

and Hindi outputs are more detail than English as all three languages identify meat but fails to identify the wooden table as in Bengali. Chinese and Japanese outputs are similar to English.

6. Conclusion

Maya drives high-quality AI content generation across languages and regions, leveraging multilingual, multimodal data to fill gaps, especially for low-resource languages. To ensure safe deployment, we rigorously curate data to remove harmful content, though some residual traces may persist. Future work will focus on refining Maya's adaptability, including testing alternative projection layers for improved cross-modal alignment and unfreezing decoder layers to optimize layer-specific fine-tuning. We plan to expand our pretrain dataset to include Bengali and Urdu translations and grow the instruction-tuning dataset to 665K examples, enhancing instruction accuracy across languages and modalities. To improve translation quality, we will tailor preambles per language. Additionally, we aim to benchmark rigorously on benchmarks such as PangeaBench [62], CVQA [45] etc. to ensure robust, diverse user support.

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Maya: An Instruction Finetuned Multilingual Multimodal Model

Supplementary Material

7. Additional Results

We compare Maya's responses with those of LLaVA-7B and GPT4 using a set of additional images. The objective is to qualitatively evaluate Maya's performance against a model of similar scale (LLaVA-7B) and a significantly larger model (GPT4, which is estimated to have trillions of parameters).The comparison includes examples of visual question answering and detailed caption generation to illustrate the strengths and differences among the models.

Visual Question Answering. Figure 11 presents a comparative analysis of model responses to a visual question identifying the brand featured in an advertisement. The conversation box displays the outputs from three models: Maya, LLaVA-7B, and GPT4. All three models successfully identify the brand as "Subway". Maya demonstrates a consistent understanding of visual content and excels at extracting OCR information, particularly when the text is clearly visible and occupies a significant portion of the image.

Figure 12 presents a comparative analysis of model responses in identifying a well known landmark. While LLaVA-7B and GPT4 correctly identify the landmark as "Diamond Head", a volcanic crater in Hawaii, Maya incorrectly identifies it as the "Seven Islands of Hawaii". This discrepancy illustrates differences in the models' accuracy and contextual understanding. The correct responses from LLaVA-7B and GPT4 demonstrate superior alignment with the visual and textual context, whereas Maya's misidentification suggests potential limitations in training on geographically specific datasets.

Caption Generation. Our example in Figure 13 illustrates how Maya generates a description to a highways-at-night scene. All models provide detailed description of the photo with different focuses. GPT4 provides precise information about the photo, including the number of lanes, the direction of the traffic, the density of the cars, the elevation of the four-lane highway, the illumination of the lights, and the presence of trees. Maya (similar to LLaVA-7B), while does not identify the number of lines, provides a vivid but detailed description of the scene.

Next, we focus on models' ability to describe a culturally diverse image in Figure 14. Here, GPT4 accurately identify the restaurant that serves this ramen, with detailed in the photo such as chashu in the ramen, the chopsticks in their paper wrap, the garnish, and the sides served with the ramen. It shows an understanding of the context. While Maya correctly identifies different elements on the table, the description is unfortunately lacks the specific culturally relevant details provided in the context.

Caption Generation in Multilingual. In Figure 15, we compare Maya's response in 10 languages in describing the image in Figure 14 in detail. Translating Hindi response to English: *The image shows a dining table with three bowls containing different Asian dishes. One of the bowls has noodles, which looks like a delicious soup. The other two bowls contain different types of vegetables, making the meal healthy and vibrant. Apart from the main dishes, there are three cups on the table, probably for beverages. A spoon is also visible, placed near the bowl of noodles, suggesting it will be used for serving the food. Overall, this scene captures a delectable and welcoming Asian dining experience*. In Chinese, Maya describes the scene as featuring *a few bowls of Asian noodles of varying kinds*, along with identifying meat, vegetables, and a spoon. It also notes the presence of *three cups possibly containing side dishes or drinks*, adding that *the whole scene creates an inviting atmosphere for people to enjoy this delicious meal together*. English translation of Spanish response: *The image shows a dining table with three bowls filled with different types of food, probably of Asian origin. One of the bowls appears to be a bowl of noodles, while the others contain a variety of vegetables. There are also three cups placed around the table, which adds to the variety of the meal. A spoon can be seen resting on the table, ready to be used to enjoy the meal. The table is filled with delicious dishes, making it a welcoming scene for those who want to enjoy a meal together*. Bengali response is similar to Spanish - they both fail to identify matcha, meat and does not identify the correct position of the spoon.

Subway

Question:
What brand is featured in this advertisement?

Maya:
The Subway brand is featured in this advertisement.
LLaVA-7B:
The brand featured in this advertisement is Subway.
GPT4:
The brand featured in this advertisement is Subway.

Figure 11. Comparison of model responses to visual questions on brand recognition.

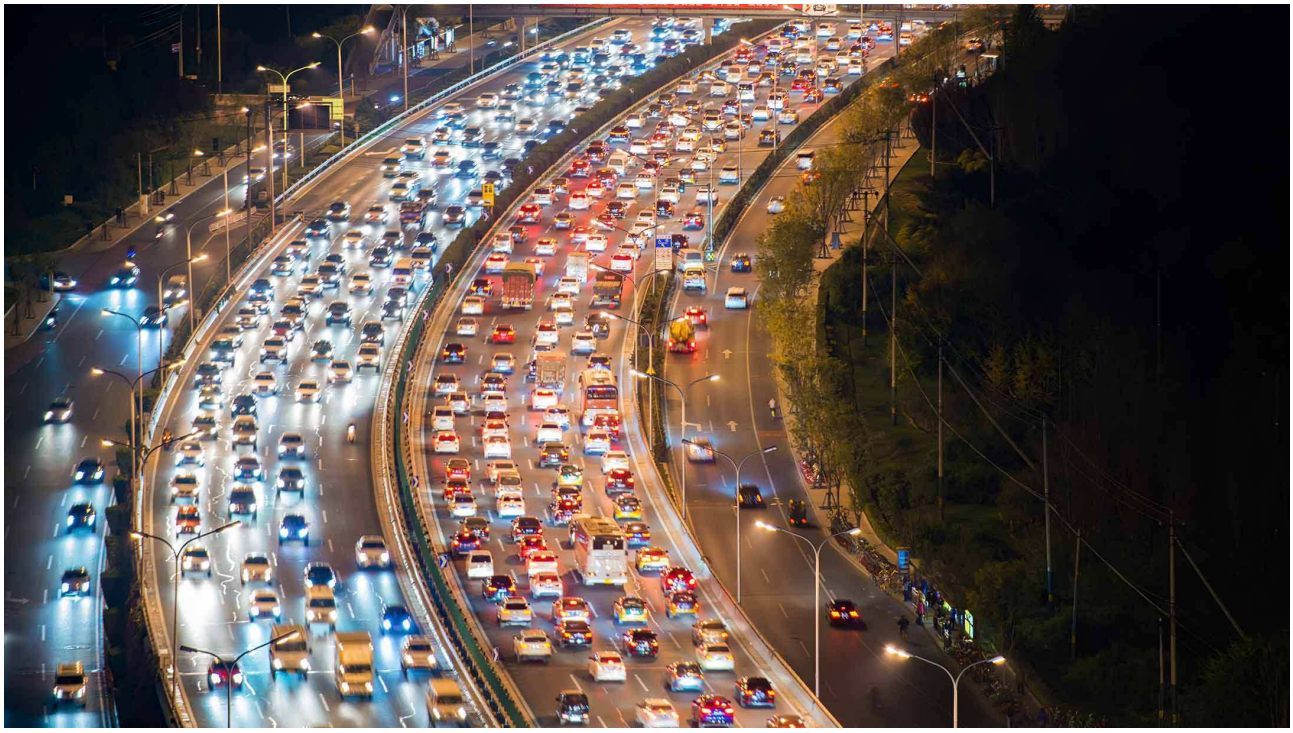
Diamond Head in Hawaii

Question:
What is the name of this famous sight in the photo?

Maya:
The famous sight in the photo is the Seven Islands of Hawaii.
LLaVA-7B:
The famous sight in the photo is the Diamond Head, which is a volcanic crater located in Hawaii.
GPT4:
The famous sight in the photo is Diamond Head.

Figure 12. Comparison of model responses to visual questions on identifying famous landmark.

Highways at Night



Question:
Describe this photo in detail.

Maya:
The image depicts a large, busy highway at night, with heavy traffic in both directions. There are numerous cars and trucks on the road, occupying almost all available lanes. The traffic appears to be moving slowly, with vehicles occupying the entire length of the highway. The scene is illuminated by the bright lights of the vehicles, as well as by the streetlights, creating a dynamic atmosphere. The highway extends into the distance, with no end in sight, highlighting the scale and congestion of this bustling transportation network.

LLaVA-7B:
The image depicts a busy highway filled with heavy traffic, including numerous cars and trucks. The highway is bustling with activity, as vehicles are seen in various positions, some moving and others waiting in line. The traffic is dense, with cars and trucks occupying almost every lane, creating a sense of congestion. The highway is surrounded by trees, adding a touch of nature to the urban landscape. The scene captures the essence of a typical day in a large city, with people commuting and traveling on the busy roads.

GPT4:
This is a top view photograph of a highway at night. The highway is divided into four sections. On the left side, there are three lanes of traffic approaching the camera and four lanes of traffic moving away from the camera. On the right side, there are four lanes of traffic approaching the camera and three lanes of traffic moving away from the camera. Most cars in the four-lane section moving away from the camera have their brake lights on, indicating a higher traffic density in the four-lane sections. Meanwhile, traffic appears to be lighter on the two three-lane sections. The four-lane highway is elevated compared to the three-lane highway, creating a visually distinct separation between the two sections. The lights alongside the highway are illuminated, providing a clear view of the road and cars. On the right side of the highway, there are trees, adding a touch of nature to the scene.

Figure 13. Comparison of model responses to visual questions on night highway traffic.

Asian Food

Question:
Describe this photo in detail.

Maya:
The image features a dining table with several bowls of food, including a large bowl of noodles and a few smaller bowls containing a variety of dishes, possibly Asian cuisine. The noodles are served in a bowl with vegetables, making it a delicious and healthy meal. There are also a couple of cups placed on the table, one near the top left corner and the other closer to the center. A spoon can be seen resting in one of the bowls, ready to be used. The table is set up with a variety of food options, making it an inviting scene for those who enjoy sharing and enjoying a meal together.

LLaVA-7B:
The image features a dining table with a variety of food items, including a large bowl of soup, a plate of noodles, and a bowl of vegetables. The soup is placed in the center of the table, while the noodles and vegetables are positioned around it. There are several bowls and cups scattered across the table, with some placed near the edges and others closer to the center. A spoon is also visible on the table, likely used for eating the soup. The table setting creates an inviting atmosphere for a meal.

GPT4:
This is a close-up photo of a meal at ICHIRAN. In the center, there is a chashu ramen bowl with a spoon. The ramen is seasoned with chili sauce and chopped scallions, and served with two pieces of chashu. To the right of the bowl, chopsticks are placed, still in their paper wrap, unopened. The ramen is accompanied by nori on the left side. On top, there are accompanying sides. From left to right, these include a bowl of orange spice (possibly garlic sauce), a plate of smoke-flavored stewed pork garnished with chopped scallions, and a cup of matcha green tea.

Figure 14. Comparison of model responses to visual questions on food details.

English	The image features a dining table with several bowls of food, including a large bowl of noodles and a few smaller bowls containing a variety of dishes, possibly Asian in nature. The noodles are served in broth and are garnished with vegetables, making it a delicious and healthy meal. There are also multiple cups placed around the table, suggesting that this is a shared meal among multiple people. A spoon can be seen resting in one of the bowls, ready to be used for serving the delicious soup. The table setting creates an inviting atmosphere for a delightful dining experience.
Spanish	La imagen muestra una mesa de comedor con tres cuencos llenos de diferentes tipos de comida, probablemente de origen asiático. Uno de los cuencos parece ser un plato de fideos, mientras que los otros contienen una variedad de verduras. También hay tres tazas colocadas alrededor de la mesa, lo que añade a la variedad de la comida. Se puede ver una cuchara descansando en la mesa, lista para ser utilizada para disfrutar de la comida. La mesa está llena de deliciosos platos, lo que la convierte en una escena acogedora para aquellos que desean disfrutar de una comida juntos.
French	L'image présente une table à manger avec plusieurs bols remplis de nourriture asiatique, notamment des nouilles, disposés dessus. Les bols sont de différentes tailles et sont placés sur toute la table. Il y a au moins trois bols de nouilles, chacun avec des ingrédients et des saveurs uniques. En plus des bols, il y a aussi deux tasses placées sur la table, l'une près du centre et l'autre plus proche du bord. Une cuillère peut être vue reposant sur l'une des assiettes, prête à être utilisée pour savourer le délicieux repas. La table est dressée pour un repas délicieux et convivial.
Chinese	这张图片展示了一张木制餐桌，上面摆放着几碗美味的亚洲面食。这些碗里有各种不同的面食，包括一些有肉和蔬菜的面条。碗的形状和大小各异，散落在桌子上。除了面食，桌子上还有三个杯子，可能装有配菜或饮料。餐桌上还放着一把勺子，准备用来吃面食。整个场景营造出一个诱人的氛围，邀请人们一起享受这顿美味的餐点。
Japanese	<画像には、さまざまなアジア料理が並べられたダイニングテーブルが映っています。テーブルの中央には、麺がたっぷり入った大きなボウルがあり、それがメインの料理のよう です。他にも、麺のボウルの近くには、野菜の料理が入ったボウルが2つあります。 ボウルに加えて、テーブルの左側にはスープが入ったカップが2つ、右側には3つ目のカ ップが置かれています。スプーンも2つあり、1つは麺のボウルの近くに置かれ、もう1つはテーブルの右側にあります。テーブルの右端には、おそらく食事を楽しむために待っ ている人のための椅子が見えます。>
Russian	Изображение показывает деревянный обеденный стол с несколькими мисками, наполненными азиатской едой, в основном лапшой. В мисках видно несколько порций лапши, каждая из которых имеет уникальный вид и подается в разных мисках. Помимо мисок, на столе расположены три чашки, расположенные в разных местах. На столе также видна ложка, вероятно, предназначенная для подачи лапши. Разнообразие блюд и мисок создает привлекательную и вкусную сцену для тех, кто собирается насладиться этим приемом пищи.
Arabic	الصورة تعرض طاولة طعام بها عدة وعاءات مليئة بأنواع مختلفة من الطعام، على الأرجح من مطبخ آسيوي. الوعاء الرئيسي هو وعاء حساء أو نوودلز، والذي يتم وضعه في وسط الطاولة. يحتوي على مكونات مثل الخضروات، مما يجعله وجبة لذيذة. بالإضافة إلى الوعاء الرئيسي، هناك ثلاثة وعاءات أخرى موضوعة حول الطاولة، تحتوي كل واحدة منها على أطباق مختلفة. يمكن رؤية ملعقة مستريحة على الطاولة، جاهزة للاستخدام في الاستمتاع بالوجبة. هناك أيضًا كوبين موضوعين بالقرب من بعضهما البعض، ربما يحتويان على مشروبات لمرافقة الوجبة.
Hindi	छवि में एक डाइनिंग टेबल दिखाई देता है जिस पर तीन बाउल्स में विभिन्न एशियाई व्यंजन हैं। इनमें से एक बाउल में नूडल्स भरे हुए हैं, जो एक स्वादिष्ट सूप की तरह दिखाई देते हैं। दूसरे दो बाउल में अलग-अलग प्रकार की सब्जियाँ हैं, जो भोजन को स्वस्थ और रंगीन बनाती हैं। मुख्य व्यंजनों के अलावा, टेबल पर तीन कप रखे गए हैं, जिनमें संभावित रूप से पेय पदार्थ हो सकते हैं। एक चम्मच भी दिखाई देता है, जो नूडल्स के बाउल के पास रखा गया है, जिससे यह सुझाव देता है कि इसे भोजन के लिए उपयोग किया जाएगा। समग्र रूप से, यह दृश्य एक स्वादिष्ट और आमंत्रणात्मक एशियाई भोजन का प्रदर्शन करता है।
Bengali	চিত্রটি একটি ডাইনিং টেবিল দেখাচ্ছে যা একটি সুস্বাদু এশিয়ান খাবারের সঙ্গে পূর্ণ। টেবিলে কয়েকটি বাটি রয়েছে, যার মধ্যে একটি বাটি নুডলস দিয়ে পূর্ণ এবং অন্যান্য বাটিতে বিভিন্ন প্রকারের খাবার রয়েছে। বাটিগুলির পাশাপাশি, টেবিলে কিছু কাপও রয়েছে, যা সম্ভবত খাবারের সঙ্গে পরিবেশন করা পানীয় ধারণ করে। টেবিলের মাঝখানে একটি চামচ রাখা হয়েছে, যা সম্ভবত নুডলস বাটিতে খাবার খেতে ব্যবহার করা হবে। টেবিলের বাম পাশে একটি বাটি দেখা যায়, যা সম্ভবত অন্য খাবার বা পরিবেশন করা পানীয় ধারণ করে। সার্বিকভাবে, দৃশ্যটি একটি আকর্ষণীয় এবং আহ্বানকর খাবার অভিজ্ঞতা প্রদর্শন করে।
Urdu	تصویر میں ایک خوراک کی میز دکھائی دیتی ہے جس پر تین باؤلز بھرے ہوئے مختلف ایشیائی ڈشز رکھے گئے ہیں۔ باؤلز میں نوڈلز اور سبزیوں کا ایک باؤل بھی ہے، جو ایک لذیذ سوپ کی طرح نظر آتا ہے۔ باؤلز کے علاوہ، میز پر تین کپ بھی رکھے گئے ہیں، جو شاید مشروبات یا اضافی اجزاء کو محفوظ کر رہے ہوں۔ میز پر ایک چمچ بھی نظر آتی ہے، جو کھانے کے لئے تیار ہے۔ یہ منظر ایک خوشگوار کھانے کا تجربہ پیش کرتی ہے، جس میں ایشیائی کھانے کے مختلف ذائقوں کا لطف اٹھایا جا سکتا ہے۔

Figure 15. Maya output for prompt (with image from Figure 9): Describe this photo in detail in {language}.